

MUHAMMAD ARGYA VITYASY

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EDUCATION

Universitas Gadjah Mada

Yogyakarta, Indonesia

Bachelor of Computer Science | Cumulative GPA: 3.85/4.00 (113/144 credits)

August 2023 – Present

- **Relevant Coursework:** Algorithms and Data Structures, Scalable Software Engineering, Software Engineering Methods, Cloud Computing, Computer Networks, Cyber Security

WORK EXPERIENCE

Software Engineer Intern | GDP Labs

February 2026 – June 2026

Enterprise AI automation platform building multi-agent LLM systems for HR, payroll, and project management workflows.

- Built **LLM-powered agents** in a multi-agent system automating enterprise HR and project management workflows, **reducing manual intervention** across 3+ workflow types.
- Designed a **compliance monitoring pipeline** that auto-generates audit summaries, eliminating manual report tracking across organizational workflows.
- Architected a **reusable, multi-tenant IAM SDK** to standardize authentication across enterprise projects, eliminating redundant setup while preventing partial-state failures via **atomic multi-provider transactions**.
- Authored **Playwright end-to-end test suites** to catch regressions and guarantee **reliable multi-agent tool execution** before deployment to production.

Backend Engineer | PT Satria Duta Raya

October 2025 – January 2026

B2B procurement platform enabling businesses to discover ranked products through an asynchronous recommendation pipeline.

- Architected a **Go/Gin backend** enforcing **6 strict domain boundaries** to prevent tight coupling and enable independent feature scaling across a B2B procurement platform.
- Built an **event-driven product ranking pipeline** on **Redis Streams** with idempotent consumers and Dead Letter Queues, decoupling heavy computation from request paths and containing failure blast radius.
- Shipped **full-stack observability** using **Prometheus and Grafana** to proactively identify performance bottlenecks and drastically **reduce Mean Time to Resolution (MTTR)** for background worker failures.

Backend Engineer | Jogja International Model United Nation 2025

April 2025 – October 2025

Annual academic simulation of the United Nations bringing together students worldwide for diplomatic debates.

- Maintained **99.9% uptime** during **500+ concurrent registration spikes** by optimizing database connection pooling and leveraging **Docker containerization** with Go and PostgreSQL.
- **Cut manual staff workload by 40%** by building a **role-based admin dashboard** that automated user verification and approval workflows for 150+ staff actions.

Backend Engineer | Organisasi Mahasiswa Ahli Teknologi Informasi UGM

March 2024 – July 2025

Student-led IT organization comprising 8 specialized divisions and 80+ members managing real-world software projects.

- Built a **reusable Go/Node.js/Docker backend framework** deployed across 4 client projects, **cutting development and deployment time by 60%**.
- Handled **3,000+ user registrations** and high-concurrency workloads with **99.9% uptime** across all deployed projects.
- **Led weekly technical discussions** for 15 engineers, facilitating deep-dives into **distributed systems, Kubernetes, and secure authentication** to drive team-wide technical growth.

PROJECTS

National University Entrance Exam Mock Platform | [GitHub](#)

- Served **200+ concurrent users** at **sub-50ms latency** across a 4-service **Go/Docker microservices architecture**.
- Processed **150+ timed exam sessions** from a 1,000-question bank with **<5% error rate** via token-based auth and a custom exam engine exposing 20+ REST APIs.
- Designed a migration plan to **Kafka-based async processing** and **Kubernetes orchestration** for future scalability.

ADDITIONAL SKILLS & INTERESTS

Languages & Frameworks: Go, Python, TypeScript, Gin-Gonic, Express, Pydantic

Data & Infrastructure: PostgreSQL, MongoDB, Redis, Kafka, Docker, Colima, Terraform, Ansible, AWS S3, MinIO

Observability & Testing: Prometheus, Grafana, Playwright

Interests: LLM-powered automation, distributed systems, cloud computing, DevSecOps, system design & scalability